Hallucination Detection in LLMs with Topological Divergence on Attention Graphs

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Abstract

Hallucination, i.e., generating factually incorrect content, remains a critical challenge for large language models (LLMs). We introduce TOHA[1], a TOpology-based HAllucination detector designed for RAG scenarios, which leverages a topological divergence metric to quantify the structural properties of graphs induced by attention maps. Examining the topological divergence between prompt and response subgraphs reveals consistent patterns: higher divergence values in specific attention heads correlate with hallucinated outputs, independent of the dataset. Extensive experiments, including evaluation on question answering and summarization tasks, show that our approach achieves state-of-the-art or competitive results on several benchmarks while requiring minimal annotated data and computational resources. Our findings suggest that analyzing the topological structure of attention matrices can serve as an efficient and robust indicator of factual reliability in LLMs.

1 Introduction

Large language models (LLMs) have progressed significantly in recent years, finding applications in various fields (Chkirbene et al., 2024). To ensure factual reliability, modern LLMs are often combined with retrieval-augmented generation (RAG) technique, which integrates relevant external knowledge from diverse databases directly into the generation process (Lewis et al., 2020).

Despite these improvements, LLMs remain prone to producing so-called *hallucinations*, i.e., content that is factually or contextually incorrect (Huang et al., 2023; Li et al., 2024). Detecting hallucinations is crucial for safe deployment of LLMs in sensitive fields since erroneous outputs may seriously harm user trust. An effective detector would therefore expand the scope of LLM applications while mitigating risks (Gao et al., 2024).

Multiple methods have been proposed to tackle this problem (Sahoo et al., 2024; Shorinwa et al., 2025). However, they often face significant practical constraints, such as the scarcity of annotated datasets (Zhang et al., 2023) required for supervised methods (Sky et al., 2024; Orgad et al., 2025), the high computational cost of generating multiple additional samples (Chen et al., 2024; Hou et al., 2025), or the inability of LLMs' output probabilities to fully represent the model's true uncertainty (Fadeeva et al., 2024; Shelmanov et al., 2025).

These challenges can be addressed by utilizing the inner states of LLMs, which are informative for the hallucination detection problem (Azaria and Mitchell, 2023; Sriramanan et al., 2024; Gekhman et al., 2025). We introduce TOHA (a TOpology-based HAllucination detector), a training-free method designed for RAG scenario that leverages the structure of LLM attention maps to identify hallucinations. Our method requires minimal annotated data while avoiding the computational overhead of multiple generations, which makes TOHA both data- and compute-efficient.

The core insight behind TOHA is that the structure of prompt-response interconnections within an LLM's attention mechanism should be indicative of hallucination presence. TOHA formalizes this idea by analyzing attention graphs — complete weighted graphs derived from LLM attention maps, the representation prior used for topological data analysis (TDA) in NLP (Kushnareva et al., 2021; Tulchinskii et al., 2023). Unlike the existing attention-based methods, which either treat all attention heads as equally important (Sriramanan et al., 2024; Binkowski et al., 2025) or ignore the geometric structure of attention maps (Sun et al., 2025; Vazhentsev et al., 2025), our method lever-

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[1]The code of the proposed method and the considered baselines is available at https://anonymous.4open.science/r/tda4hallucinations-C449

ages a small set of attention heads selected by a topological criterion designed to identify hallucinations.

Specifically, TOHA is based on the $\mathrm{MTop\text{-}Div}_G(R, P)$, our adaptation of Manifold Topology Divergence (Barannikov et al., 2021) for the graph setting, which quantifies the dissimilarity between the prompt (P) and response (R) token sets in the attention graph. We demonstrate that this score not only reflects the geometry but also measures informational novelty of the response in relation to the prompt, thus being well-suited for hallucination detection in RAG scenarios.

Through analysis of divergence values across attention heads, we identified a subset of heads that consistently yield greater scores for hallucinated samples (see Figure 2), irrespective of the dataset. TOHA's final hallucination score is the average $\mathrm{MTop\text{-}Div}_G(R, P)$ value from these "hallucination-aware" heads. We discover that some of these heads are associated with copying behavior, which is aligned with prior findings (Sun et al., 2025).

Our main contributions can be summarized as follows:

- We introduce TOHA, a training-free framework for hallucination detection that leverages the topology of LLM attention graphs. Our method operates up to an order of magnitude faster than methods of comparable quality and requires minimal annotated data.
- Central to TOHA is $\mathrm{MTop\text{-}Div}_G(R, P)$, an adaptation of Manifold Topology Divergence for graphs, which quantifies the topological dissimilarity between prompt and response sets of tokens. We demonstrate that this score characterizes the novelty of the response with respect to the prompt, making it highly effective for detecting hallucinations in RAG systems.
- By analyzing the proposed score, we discover the existence of "hallucination-aware" attention heads, which consistently yield greater divergence values for hallucinated samples across different datasets. This finding ensures TOHA's efficiency and solid cross-domain transferability: averaging $\mathrm{MTop\text{-}Div}_G(R, P)$ for just ten heads is enough for robust hallucination detection.
- Our experiments show that TOHA consistently matches or exceeds state-of-the-art performance across multiple benchmarks when applied to open-source LLMs of varying scales, including 7B and 13B parameters.

2 Background

2.1 Attention Map as a Weighted Graph

Modern LLMs are mainly based on the self-attention mechanism, introduced in (Vaswani et al., 2017). Let $X \in \mathbb{R}^{n\times d}$ be a matrix consisting of d-dimensional representations of n tokens, W_Q, W_K, $W_V \in \mathbb{R}^{d\times d}$ be trainable projection matrices. Given a set of queries $Q = XW_Q \in \mathbb{R}^{n\times d}$, a set of keys $K = XW_K \in \mathbb{R}^{n\times d}$, and corresponding values $V = XW_V \in \mathbb{R}^{n\times d}$, the attention mechanism calculates a weighted sum of the values as follows:

$$\mathrm{Attention}(Q, K, V) = W(Q, K)V, \qquad (1)$$

where $W(Q, K)$ is an attention map

$$\mathrm{W}(Q, K) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d}}\right), \qquad (2)$$

and each entry $w_{ij} = W_{ij}(Q, K)$ captures how strongly token i attends to token j, with greater values indicating closer relationship. We consider decoder-only LLMs, for which the attention maps are lower-triangular.

An attention map can be reframed as a complete undirected weight graph G with edge weights $1 - w_{ij}$ to represent pseudo-distances between tokens. We call G an *attention graph*. It naturally partitions into prompt (P) and response (R) tokens (see Figure 1b). We analyze the topological relationships between these node subsets, which, as we assume, should be indicative of hallucinations in the RAG scenario.

2.2 Manifold Topology Divergence

The Manifold Topology Divergence, or MTop-Div, was proposed in (Barannikov et al., 2021). $\mathrm{MTop\text{-}Div}(M, N)$ quantifies the difference between two data manifolds $\mathcal{M}$ and $\mathcal{N}$, approximated by point clouds M and N. It is computed as the sum of interval lengths in the $\mathrm{Cross\text{-}Barcode}(M, N)$, a set of intervals representing topological features distinguishing N from $M \cup N$. A larger divergence indicates a greater topological difference between the manifolds. For more details on topological data analysis, see Appendix C.

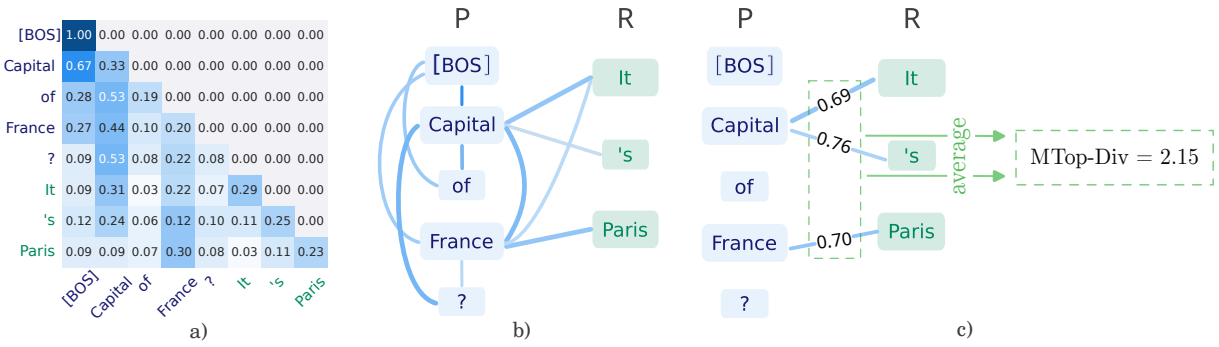


Figure 1: a) An attention map. Blue and green denotes the prompt and response tokens, respectively. b) The corresponding attention graph G. Prompt tokens P are located on the left, response tokens R — on the right. To keep figure neat, we only plot the edges with an attention score of no less than 0.15. c) The minimum spanning forest attaching R to P and the corresponding MTop-Div value.

3 Method

This section introduces the $\text{MTop-Div}_G(R, P)$ score, which is designed to quantify the topological divergence between the prompt and response subgraphs in attention maps. We demonstrate that not only can it be interpreted as a geometric characteristic of the attention graph, but also as an information-theoretic measure of the novelty of the response in relation to the prompt. With $\text{MTop-Div}_G(R, P)$, we identify "hallucination-aware" heads that consistently separate hallucinated samples from grounded, irrespective of dataset. This finding allows us to formulate the TOHA algorithm (Algorithm 1), which computes a final hallucination score by averaging the divergence values from these specific heads.

3.1 MTop-Div for Attention Graphs: Definition

Let R and P be the response and prompt vertex sets in an attention graph G. After zeroing edge weights between the P nodes, we compute the 0-th order homology barcode $\mathcal{B}_0$ of the Vietoris-Rips simplicial complex of the modified graph. Essentially, this barcode tracks the evolution of connected components as we threshold the graph edges (see Appendix C for details).

Our proposed topological divergence is a sum of lengths of $\mathcal{B}_0$ intervals:

$$\text{MTop-Div}_G(R, P) = \sum_{[b_i, d_i] \in \mathcal{B}_0} |d_i - b_i|.$$

This score can be interpreted from two perspectives: geometric (as the length of a minimal spanning forest in the attention graph) and information-theoretic (as a measure of the response novelty in the space induced by query-key matrices). We explore these two interpretations in the following subsections.

3.2 $\text{MTop-Div}_G(R, P)$ and Geometry of Attention Graph

Formally, we can prove the following property (see proof in Appendix C):

Proposition 3.1. *Consider an attention graph G with vertex set V_G and its complementary vertex subsets P, R, where $P \cup R = V_G$ and $P \cap R = \varnothing$. $\text{MTop-Div}(R, P)$ value equals the length of the minimal spanning forest (MSF) attaching R to P.*

This proposition's geometric meaning is illustrated in Figure 1. While the connections between prompt nodes (Figure 1b) are semantically and syntactically meaningful, we hypothesize they primarily introduce noise for hallucination detection (see Section 4.4 for the corresponding experiment). After setting these distances to zero, we construct an MSF on the modified graph (Figure 1c). The sum of the lengths of the remaining edges in this MSF is precisely the value of $\text{MTop-Div}_G(R, P)$.

3.3 $\text{MTop-Div}_G(R, P)$ and Perplexity in the Query-Key Space

Consider an attention graph G. It can be interpreted as a non-metric* space, with "distances" induced by Q and K matrices from the attention mechanism.

In order to quantify some notion of entropy in this non-metric space, the most intuitive way is to measure the MST length (Müller et al., 2012) as

*The $Q^T K$ scalar product from self-attention cannot be formally interpreted as a metric: for one, it does not adhere to the triangle's inequality.

a proxy for token dispersion. More precisely, the entropy estimate given the MST of length L on points $\mathcal{X}$ is the following:

$$H_{MST}(\mathcal{X}) = d \log L - (d-1) \log n + \log \beta_d,$$

where $|\mathcal{X}| = n$, d is the intrinsic dimensionality of the data, and β_d is some data-independent constant.

We can show that

$$\text{MTop-Div}_G(R, P) \geq L_{MST}(R \cup P) - L_{MST}(P),$$

where the first term corresponds to the MST length of the complete attention graph G, and the second to the length of the MST spanning the prompt tokens P. Therefore, our score has the following lower bound:

$$\text{MTop-Div}_G(R, P) \geq \alpha 2^{H(P \cup R)} - \beta 2^{H(P)},$$

$\alpha, \beta > 0$, or, in terms of the perplexity function,

$$\text{MTop-Div}_G(R, P) \geq \alpha \text{PPL}(\text{P} \cup \text{R}) - \beta \text{PPL}(\text{R}).$$

Perplexity measures how "surprising" a piece of data is. Therefore, the difference between the perplexity values quantifies the novelty of the response with respect to the prompt: greater values suggest that new information is present, which, in the RAG scenario, is considered to be a hallucination. Thus, our $\text{MTop-Div}_G(R, P)$ should be an effective statistic for identifying hallucinated responses.

3.4 Hallucination-Aware Heads

Denote by h_{ij} the j-th attention head from the layer i. For the specific data sample s and head h_{ij}, let G^s_{ij} be the corresponding attention graph, P^s_{ij}, R^s_{ij} — its prompt and response vertex subsets.

We examined typical values of the average distance between hallucinated and grounded training examples for different heads and layers:

$$\Delta_{ij} = \frac{1}{|S_{\text{hallu}}|} \sum_{s \in S_{\text{hallu}}} d_{ij}(s) - \frac{1}{|S_{\text{gr}}|} \sum_{s \in S_{\text{gr}}} d_{ij}(s), \tag{3}$$

where S_{hallu} stands for all hallucinated samples from the training set, S_{gr} stands for all grounded training samples, and

$$d_{ij}(s) = \frac{1}{|R^s_{ij}|} \text{MTop-Div}_{G^s_{ij}}(R^s_{ij}, P^s_{ij}).$$

Figure 2 displays sample differences Δ_{ij} across three datasets, with each marker representing some attention head. We discovered that the same four (for Mistral-7B) and three (for Llama-2-7B) heads, highlighted in pink, demonstrate similar behaviour across the datasets: they consistently appear in the upper-right corner, indicating strong separation between hallucinated and grounded samples, irrespective of the dataset. This finding suggests that there are specific attention heads somehow "aware" of the presence of hallucination, which is captured by the proposed $\text{MTop-Div}_G(R, P)$.

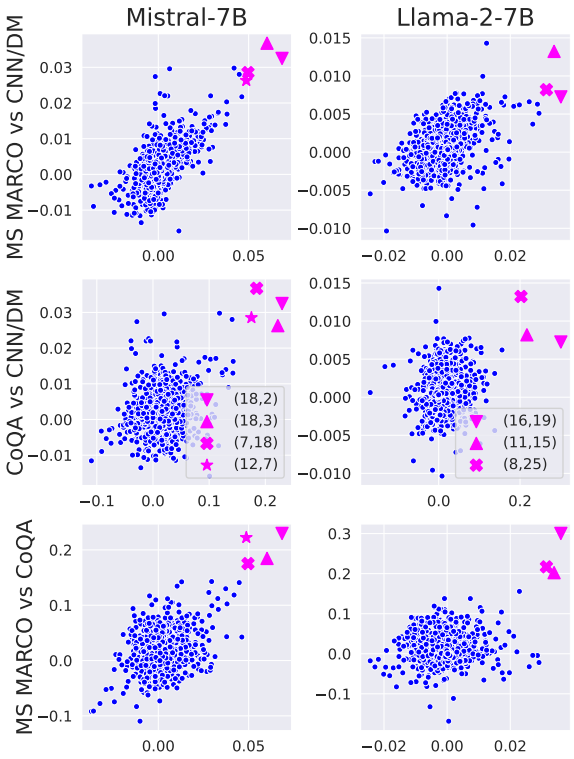


Figure 2: Δ_{ij} values for ij-th heads. Vertical axis corresponds to the difference on dataset (B), horizontal — to the one on dataset (A). The heads that separate samples best are highlighted in pink; their (layer, head) positions are reflected in the legend.

3.5 TOHA

The existence of "hallucination-aware" attention heads underlies our method, which is detailed in Algorithm 1. TOHA uses two annotated probe sets (S_h for hallucinations, S_g for grounded samples) to rank attention heads by their separation capability Δ_{ij} and select the top N_{opt} ones with the combined probe size $|S_h \cup S_g|$ kept small (see Figure 3). During testing, the final hallucination score for test samples T equals the average topological divergence from the selected N_{opt} heads. For computational efficiency, we limit N_{opt} to a maximum of $N_{\max} = 10$ in all experiments.

Algorithm 1 TOHA Algorithm

Require: $d_{ij}(s)$, S_h, S_g, $V = S_h \cup S_g$, T, $N_{\max}$
Ensure: Hallucination scores $\{p_s\}$ for $s \in T$

function HEADSSELECTION
  for each head h_{ij} **do**
    Calculate Δ_{ij} according to eq. (3)
  end for
  $H \leftarrow$ sort all heads by Δ_{ij} (descending)
  $N_{\text{opt}}, \text{AUROC}_{\max} \leftarrow 1, 0$
  Initialize $p_s \leftarrow 0$ for all $s \in V$
  for $N = 1$ **to** $N_{\max}$ **do**
    for each $s \in V$ **do**
      $p_s \leftarrow \frac{N-1}{N} p_s + \frac{1}{N} d_{h_N}(s)$
    end for
    auroc $\leftarrow \text{AUROC}(\{y_s\}_{s\in V}, \{p_s\}_{s\in V})$
    if auroc $> \text{AUROC}_{\max}$ **then**
      $\text{AUROC}_{\max} \leftarrow$ auroc; $N_{\text{opt}} \leftarrow N$
    end if
  end for
  return N_{opt}
end function

function PREDICTION(N_{opt})
  for each $s \in T$ **do**
    $p_s \leftarrow \frac{1}{N_{\text{opt}}} \sum_{i=1}^{N_{\text{opt}}} d_{h_i}(s)$
  end for
  return $\{p_s\}$
end function

4 Results

4.1 Experiment Setting

Datasets and models. We consider five tasks that evaluate question answering and summarization abilities of LLMs: two benchmarks of RAGTruth (Niu et al., 2023) — long-form QA dataset MS MARCO and summarization dataset CNN/DM; conversational QA dataset CoQA (Reddy et al., 2019), reading comprehension dataset SQuAD (Rajpurkar et al., 2016), and extreme summarization dataset XSum (Narayan et al., 2018). For more details, see Appendix E.

We employ five popularly adopted open-source LLMs with accessible inner states: LLaMA-2-7B-chat, LLaMA-2-13B-chat, LLaMA-3.1-8B-Instruct, Mistral-7B-Instruct-v0.1, and Qwen2.5-7B-Instruct. As the RAGTruth dataset does not contain responses for LLaMA-3.1-8B and Qwen-2.5-7B, we conducted experiments on SQuAD, CoQA, and XSum for these models.

Baselines. We compare TOHA with a comprehensive set of eight baselines: uncertainty-based perplexity (Ren et al., 2023) and max entropy (Fadeeva et al., 2024); inner states-based ReDEEP (Sun et al., 2025), HaloScope (Du et al., 2024), and LLM-Check (Sriramanan et al., 2024); consistency-based semantic entropy (Farquhar et al., 2024), EigenScore (Chen et al., 2024), and SelfCheckGPT (Manakul et al., 2024).

Implementation details. The reported results are averaged over 5 runs with different data splits, using test sets comprising 25% of the data and a fixed validation set size of 100, following the methodology of HaloScope (Du et al., 2024). To ensure a fair comparison, we consider two implementations of the latter: a standard setting with 20 generations (see Tables 5-6), and an efficiency-comparable one with minimal number of generations — 1 for SelfCheckGPT, 5 for semantic entropy and EigenScore (Tables 1-2). Appendix F provides additional information on implementation details.

4.2 Results

Main results. The results of our experiments are provided in Tables 1–2. We evaluate TOHA against state-of-the-art hallucination detection methods and demonstrate its competitive performance TOHA significantly outperforms uncertainty-based baselines and matches the quality of consistency-based approaches, achieving notable improvements of 11.7% on the challenging long-form QA MS MARCO dataset for Mistral-7B and 21.6% on the conversational QA dataset CoQA for LLama-2-7B.

To evaluate TOHA's robustness to data distribution change, we conducted transfer experiments on Mistral-7B, see Figure 3(c) for the results. TOHA possesses strong transferability: for the XSum and CNN/DM datasets, performance in transfer settings fall within the standard deviation of the method (see Table 9), while remaining competitive on the other datasets (Table 1).

Evaluation on a multi-hop dataset. To validate TOHA in a more realistic setting, we consider an additional experiment using the HotpotQA (Yang et al., 2018) dataset. It consists of questions that require knowledge from multiple supporting documents — much like real-world queries, which also rarely rely on a single source of information. The results are provided in Table 3. TOHA demon-

Table 1: ROC AUC (↑) of hallucination detection techniques for three LLMs. The best results for each model are highlighted in **bold**, and the second best are underlined.

Method	Single generation	MS MARCO	CNN/DM + Recent News	CoQA	SQuAD	XSum
Mistral-7B						
SelfCheckGPT [1]	✗	0.63 ± 0.04	0.51 ± 0.04	0.86 ± 0.02	0.71 ± 0.04	0.66 ± 0.04
Semantic entropy [2]	✗	0.54 ± 0.03	0.51 ± 0.04	0.83 ± 0.02	0.70 ± 0.03	0.56 ± 0.03
EigenScore [3]	✗	0.54 ± 0.04	0.50 ± 0.06	0.74 ± 0.02	0.71 ± 0.04	0.58 ± 0.04
HaloScope [4]	✓	0.57 ± 0.08	0.51 ± 0.10	0.62 ± 0.08	0.92 ± 0.07	0.62 ± 0.02
LLM-Check [5]	✓	0.49 ± 0.03	0.49 ± 0.03	0.60 ± 0.01	0.50 ± 0.03	0.58 ± 0.04
Perplexity [6]	✓	0.45 ± 0.01	0.54 ± 0.02	0.54 ± 0.03	0.81 ± 0.05	0.54 ± 0.06
Max entropy [7]	✓	0.68 ± 0.04	**0.60 ± 0.07**	0.73 ± 0.00	0.75 ± 0.05	**0.71 ± 0.02**
ReDEEP [8]	✓	0.54 ± 0.02	0.47 ± 0.06	0.59 ± 0.03	0.45 ± 0.05	0.63 ± 0.04
TOHA (ours)	✓	**0.76 ± 0.04**	**0.60 ± 0.09**	**0.89 ± 0.01**	**0.96 ± 0.01**	0.66 ± 0.05
LLama-2-7B						
SelfCheckGPT [1]	✗	0.59 ± 0.03	**0.60 ± 0.03**	0.66 ± 0.03	0.57 ± 0.03	0.64 ± 0.05
Semantic entropy [2]	✗	0.53 ± 0.03	0.51 ± 0.03	0.76 ± 0.01	0.73 ± 0.03	0.61 ± 0.04
EigenScore [3]	✗	0.55 ± 0.03	0.53 ± 0.04	0.61 ± 0.03	0.75 ± 0.02	0.63 ± 0.02
HaloScope [4]	✓	0.51 ± 0.05	0.48 ± 0.05	0.61 ± 0.04	0.67 ± 0.04	0.57 ± 0.07
LLM-Check [5]	✓	0.44 ± 0.02	0.49 ± 0.06	0.60 ± 0.03	0.49 ± 0.01	0.61 ± 0.01
Perplexity [6]	✓	0.54 ± 0.04	0.44 ± 0.03	0.74 ± 0.02	0.46 ± 0.03	0.56 ± 0.09
Max entropy [7]	✓	**0.65 ± 0.04**	0.59 ± 0.06	0.65 ± 0.03	0.73 ± 0.04	0.56 ± 0.03
ReDEEP [8]	✓	0.54 ± 0.04	0.52 ± 0.04	0.72 ± 0.04	0.42 ± 0.08	0.54 ± 0.06
TOHA (ours)	✓	**0.65 ± 0.02**	0.56 ± 0.02	**0.90 ± 0.01**	**0.87 ± 0.04**	**0.68 ± 0.05**
LLaMA-2-13B						
SelfCheckGPT [1]	✗	0.58 ± 0.04	**0.58 ± 0.05**	0.77 ± 0.02	0.64 ± 0.03	0.60 ± 0.04
Semantic entropy [2]	✗	0.57 ± 0.04	0.54 ± 0.03	0.76 ± 0.04	0.65 ± 0.03	0.60 ± 0.03
EigenScore [3]	✗	0.56 ± 0.04	0.47 ± 0.04	0.57 ± 0.03	0.57 ± 0.02	0.52 ± 0.06
HaloScope [4]	✓	0.54 ± 0.09	0.51 ± 0.04	0.57 ± 0.03	0.55 ± 0.02	0.55 ± 0.07
LLM-Check [5]	✓	0.49 ± 0.06	0.56 ± 0.05	0.57 ± 0.02	0.57 ± 0.07	0.57 ± 0.07
Perplexity [6]	✓	0.54 ± 0.04	0.46 ± 0.07	0.62 ± 0.03	0.45 ± 0.02	0.49 ± 0.05
Max entropy [7]	✓	0.62 ± 0.03	0.53 ± 0.06	0.66 ± 0.03	0.78 ± 0.02	0.59 ± 0.04
ReDEEP [8]	✓	0.62 ± 0.06	0.48 ± 0.05	0.73 ± 0.02	0.48 ± 0.07	0.58 ± 0.08
TOHA (ours)	✓	**0.67 ± 0.04**	0.56 ± 0.05	**0.92 ± 0.02**	**0.88 ± 0.05**	**0.66 ± 0.03**

strates superior performance to baselines, confirming its effectiveness "in the wild".

Efficiency. Figure 4(a) shows that TOHA is approximately seven times faster than SelfCheckGPT with a *single* additional generation. Given that SelfCheckGPT typically requires 10-20 generations, this makes TOHA over 70 times faster in practice. Furthermore, TOHA's runtime is close to the lightweight entropy baseline; with a more efficient low-level implementation, it has the potential to approach the cost of a single forward pass while achieving higher accuracy than other inexpensive alternatives.

4.3 Analysis of Hallucination-Aware Attention Heads

Attention patterns and copying behaviour. For hallucination-aware heads, we analyzed minimal spanning forest (MSF) patterns distinguishing hallucinated and grounded samples. A key finding is that for these heads, hallucinated samples frequently exhibit strong attention to the first token, whereas grounded samples tend to attend to the first token less. An example is provided in Figure 5.

Table 2: ROC AUC (↑) of hallucination detection techniques. The best results for each model are highlighted in **bold**, and the second best are underlined.

Method	Single gen.	CoQA	SQuAD	XSum
LLaMA-3.1-8B				
SelfCheckGPT [1]	✗	**0.91 ± 0.01**	0.65 ± 0.05	**0.68 ± 0.05**
Semantic entropy [2]	✗	0.78 ± 0.02	0.55 ± 0.07	0.47 ± 0.04
EigenScore [3]	✗	0.80 ± 0.02	0.45 ± 0.05	0.50 ± 0.03
HaloScope [4]	✓	0.67 ± 0.08	0.84 ± 0.07	0.55 ± 0.06
LLM-Check [5]	✓	0.54 ± 0.06	0.49 ± 0.05	0.52 ± 0.04
Perplexity [6]	✓	0.51 ± 0.05	0.68 ± 0.02	0.65 ± 0.03
Max entropy [7]	✓	0.82 ± 0.03	0.60 ± 0.02	0.65 ± 0.03
ReDEEP [8]	✓	0.58 ± 0.08	0.39 ± 0.04	0.62 ± 0.06
TOHA (ours)	✓	0.84 ± 0.01	**0.87 ± 0.03**	0.65 ± 0.05
Qwen2.5-7B				
SelfCheckGPT [1]	✗	0.69 ± 0.01	0.74 ± 0.02	**0.69 ± 0.02**
Semantic entropy [2]	✗	0.76 ± 0.04	0.57 ± 0.04	0.64 ± 0.03
EigenScore [3]	✗	0.78 ± 0.03	0.55 ± 0.04	0.55 ± 0.04
HaloScope [4]	✓	0.66 ± 0.13	0.75 ± 0.04	0.57 ± 0.07
LLM-Check [5]	✓	0.53 ± 0.07	0.54 ± 0.06	0.54 ± 0.02
Perplexity [6]	✓	0.39 ± 0.03	0.65 ± 0.03	0.53 ± 0.05
Max entropy [7]	✓	**0.85 ± 0.02**	0.47 ± 0.05	0.60 ± 0.06
ReDEEP [8]	✓	0.37 ± 0.10	0.56 ± 0.03	0.67 ± 0.03
TOHA (ours)	✓	0.79 ± 0.05	**0.77 ± 0.02**	**0.69 ± 0.03**

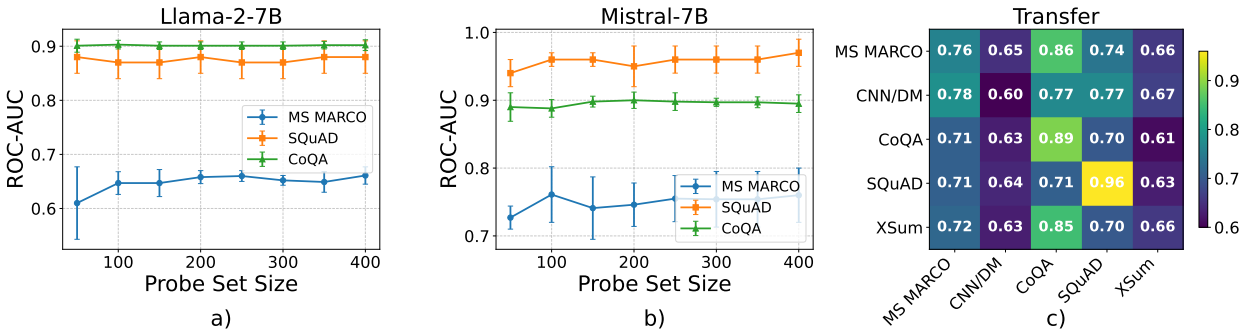


Figure 3: (a)-(b): Detection quality dependence on the size of a probe set, models: Mistral-7B (left), LLama-2-7B (right). (c) Generalizability between the datasets, model: Mistral-7B. The vertical axis corresponds to the origin of the probe set, the horizontal axis to the test dataset.

Method	Single gen.	Mistral-7B	Llama-2-13B
SelfCheckGPT	✗	0.70 ± 0.06	0.63 ± 0.04
Semantic entropy	✗	0.70 ± 0.05	0.70 ± 0.04
EigenScore	✗	0.68 ± 0.04	0.67 ± 0.04
HaloScope	✓	0.60 ± 0.06	0.50 ± 0.01
LLM-Check	✓	0.48 ± 0.03	0.56 ± 0.04
Perplexity	✓	0.55 ± 0.06	0.49 ± 0.04
Max entropy	✓	0.62 ± 0.04	0.69 ± 0.05
ReDEEP	✓	0.49 ± 0.04	0.62 ± 0.05
TOHA (ours)	✓	**0.71 ± 0.08**	**0.80 ± 0.03**

Table 3: ROC-AUC (↑) values on the HotpotQA dataset. Best results are highlighted in **bold**, and the second best are underlined.

Since this pattern is a known behavior of *token induction*, or *copying* heads — which default to the first token when unable to find previous occurrences of the current token in the context (Elhage et al., 2021) — we decided to explore the relationship between these special heads and hallucination-aware ones.

To investigate this link, we ranked all attention heads in Llama-2-7B and Mistral-7B based on their copying scores, following the method of (Feucht et al., 2025). For the subset of heads frequently identified by TOHA (appearing in $\geq 20\%$ of runs across all datasets), their copying ranks were recorded. The results, shown in Figure 6, reveal that hallucination-aware heads are often also among the model's top-25 copiers, a finding that aligns with prior work (Sun et al., 2025).

4.4 Ablation Studies

Why zero out the distances between prompt tokens? We hypothesize that the connections within the prompt contribute little to hallucination detection; therefore, our method is designed to disregard them. To validate this architectural choice, we considered an ablation study, where

Table 4: ROC-AUC (↑) of Algorithm 1 with MST length of the complete graph vs $\mathrm{MTop\text{-}Div}_G(R, P)$. Best results are highlighted in **bold**.

Dataset	MST length	$\mathrm{MTop\text{-}Div}_G(R, P)$
Mistral-7B		
CoQA	0.75 ± 0.03	$\mathbf{0.90 \pm 0.01}$
MS MARCO	0.38 ± 0.03	$\mathbf{0.65 \pm 0.02}$
LLama-2-7B		
CoQA	0.60 ± 0.03	$\mathbf{0.89 \pm 0.01}$
MS MARCO	0.37 ± 0.02	$\mathbf{0.76 \pm 0.04}$

$\mathrm{MTop\text{-}Div}_G(R, P)$ was replaced by the MST length of the complete graph in Algorithm 1. The results showing the effectiveness of the proposed approach are in Table 4.

What about other attention map-based features? To demonstrate that our topology-based approach offers an advantage over conventional attention map features, we compared them to MTop-Div in the supervised setting. Table 8 from Appendix B shows that a classifier trained on $\mathrm{MTop\text{-}Div}_G(R, P)$ achieves the best performance, confirming that our score captures unique patterns in an attention map that standard approaches miss.

How large should the probe sets be? To evaluate TOHA's sensitivity to probe set size, we conducted a sensitivity study. The results from Figure 3 confirm TOHA's robustness: even with just 50 samples, the performance drop remains minor with stable results as the probe set size increases.

How many attention heads do we need? To evaluate the sensitivity of our method to the hyperparameter N_{max}, we performed an ablation study for values from 1 to 10. As shown in Figure 4, TOHA achieves strong detection performance even

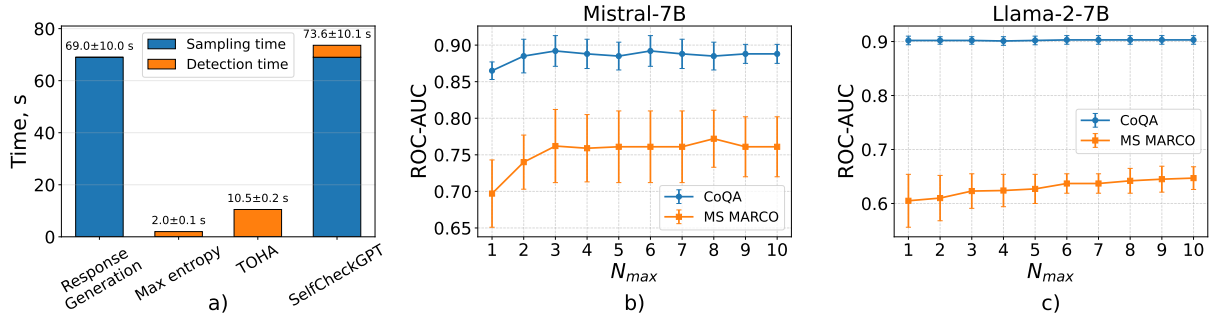


Figure 4: a): Inference time comparison (seconds) for various methods evaluated on 16 MS MARCO samples using Mistral-7B. SelfCheckGPT measurement includes one additional generated answer per sample. b)-c): ROC-AUC performance of TOHA across different numbers of selected attention heads (N_{max}) on Mistral-7B and Llama-2-7B.

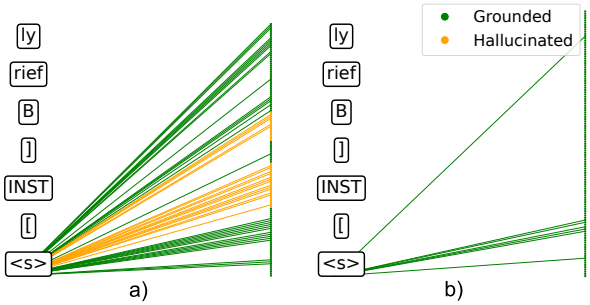


Figure 5: Attention to the first token (<s> in this example) for (a) a hallucinated generation and (b) a grounded one. Green highlights edges and nodes corresponding to grounded tokens, while yellow indicates hallucinated tokens. Model: Mistral-7B.

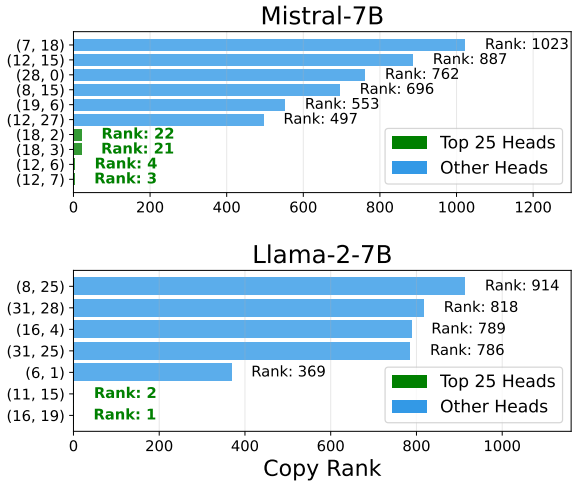


Figure 6: Copying ranks of hallucination-aware attention heads (lower ranks indicate stronger copying behaviour). The row labels (X, Y) correspond to X-th layer and Y-th head.

when $N_{\max} = 1$, which underscores the effectiveness of our topological approach.

5 Conclusion

This paper introduces TOHA, a novel hallucination detection method based on the topological structure of attention maps. Central to TOHA is the $\mathrm{MTop\text{-}Div}_G(R, P)$, our adaptation of Manifold Topology Divergence (Barannikov et al., 2021) for the graph setting, which, as we demonstrate, serves as a measure of the response novelty in relation to the prompt. This property makes the proposed divergence well-suited for hallucination detection in RAG scenarios.

By analyzing the divergence values, we identified a subset of "hallucination-aware" attention heads that reliably distinguish hallucinated and grounded samples, irrespective of the dataset. TOHA computes the final hallucination scores by averaging the topological divergences from these heads. Further investigation reveals that some of these heads are associated with copying behavior, which aligns with prior work (Sun et al., 2025).

Extensive experiments show that TOHA is a robust alternative to existing approaches, matching or surpassing state-of-the-art baselines. Moreover, our method is both data- and compute-efficient: just 50 annotated samples suffice for reliable detection, and inference runs several times faster than comparable methods of similar quality. Crucially, we validate TOHA's transferability, demonstrating its robustness to shifts in data distribution — a key advantage for real-world deployment, where LLM inputs are far more diverse and complex than benchmark examples.

In summary, TOHA delivers state-of-the-art detection performance while combining efficiency and solid generalizability, making it particularly well-suited for practical applications.

Limitations

While TOHA demonstrates strong performance and efficiency, several limitations warrant discussion.

Model-specific dependencies. TOHA's effectiveness relies on identifying "hallucination-aware" attention heads, which may vary across LLM architectures. While our experiments cover popular open-source models (e.g., LLaMA, Mistral), further validation is needed for proprietary or larger models (e.g., GPT-4, Claude).

Multimodal extensions. The current framework operates solely on text. Adapting TOHA to multimodal settings (e.g., vision-language models) would require redefining attention graphs across heterogeneous data modalities.

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A Related Works

Hallucination detection methods. The problem of hallucinations in LLMs has attracted significant attention recently (Zhang et al., 2023; Huang et al., 2023; Wang et al., 2024). Consistency-based methods (Manakul et al., 2024; Chen et al., 2024; Qiu and Miikkulainen, 2024; Nikitin et al., 2024; Hou et al., 2025) that use the diversity of multiple LLM responses as a hallucination score offer robust detection but impose significant computational overhead. Surface-level techniques like perplexity and logit entropy (Fadeeva et al., 2024; Malinin and Gales, 2021) analyze model confidence directly from output distributions — efficient but limited in detection capability as they neglect the model's rich internal representations (Gekhman et al., 2025). Hidden states-based classifiers (Azaria and Mitchell, 2023; Sky et al., 2024; Zhou et al., 2025) require extensive annotated datasets, which are scarce in the public domain (Zhang et al., 2023), and suffer from poor task transferability (Sky et al., 2024). While HaloScope (Du et al., 2024) partially addresses the data scarcity issue by leveraging unlabeled "in-the-wild" generations, it still demands a large volume of unannotated model outputs to achieve strong performance.

Attention map-based methods represent a promising yet underdeveloped direction. Current techniques either (i) rely on large annotated datasets (e.g., Lookback Lens (Chuang et al., 2024), LapEigVals (Binkowski et al., 2025)), (ii) exploit only simplistic attention graph properties like self-loop weights (LLM-Check (Sriramanan et al., 2024)), or (iii) disregard attention map geometry entirely, using mechanistic scores instead (ReDEEP (Sun et al., 2025)). This leaves a critical research gap: training-free methods that fully leverage the rich structural information encoded in attention graphs remain underexplored.

Topological Data Analysis (TDA) in NLP. Topological Data Analysis (TDA) is a mathematical framework for extracting multi-scale structural patterns from data using principles from topology and computational geometry (Chazal and Michel, 2017; Hensel et al., 2021). Recent years have seen growing interest in applying TDA to natural language processing (NLP) tasks to study textual structural properties (Uchendu and Le, 2024). For example, (Tulchinskii et al., 2024) leveraged persistent homology to estimate the intrinsic dimensionality of CLS embeddings for detecting machine-generated text. Other work has demonstrated the utility of topological features derived from transformer attention matrices — treated as weighted graphs — for diverse NLP applications. These include uncertainty quantification (Kostenok et al., 2023) and grammatical acceptability classification (Proskurina et al., 2023), where topological features extracted from the attention graphs were used as input to train auxiliary classifiers.

B Additional Experiment Results

B.1 Multiple generations-based methods

To provide a complete comparison, we considered an implementation of consistency-based methods with 20 additional generations. The results are demonstrated in Tables 5-6. We can see that TOHA still remains superior to the baselines, achieving top performance in most experiments.

B.2 Comparison to ReDEEP

For a comprehensive comparison with ReDEEP (Sun et al., 2025), which was evaluated on the entire RAGTruth dataset in the original paper, we conducted this evaluation of TOHA under the same conditions. As shown in Table 7, TOHA not only outperforms ReDEEP but also demonstrates greater robustness to data distribution shifts, exhibiting less significant performance degradation on separate benchmarks of RAGTruth (Tables 1-2).

B.3 Alternative attention map-based features for hallucination detection

During preliminary experiments for an attention map-based hallucination detector, we evaluated a range of topological and traditional features. As standard topological features, we employed barcode-based features, such as the sum of bar lengths in persistence diagrams, and naive topological features, including the average vertex degree in attention graphs (Kushnareva et al., 2021). As for traditional attention-based features, we used sparsity ratio, attention entropy, and spectral norm (Kobayashi et al., 2020; Vig and Belinkov, 2019; Ji et al., 2021). We also considered Wasserstein distances between the persistent diagrams (Edelsbrunner and Harer, 2010) of the prompt and response subgraphs as an alternative

Table 5: ROC AUC (↑) of multiple generations-based methods (with 20 additional samples) and TOHA. The best results for each model are highlighted in **bold**, and the second best are underlined.

Method	Single generation	MS MARCO	CNN/DM + Recent News	CoQA	SQuAD	XSum
Mistral-7B						
SelfCheckGPT [1]	✗	0.67 ± 0.03	0.59 ± 0.04	**0.93 ± 0.01**	0.83 ± 0.02	**0.71 ± 0.04**
Semantic entropy [2]	✗	0.53 ± 0.03	0.52 ± 0.03	0.86 ± 0.01	0.74 ± 0.01	0.63 ± 0.02
EigenScore [3]	✗	0.49 ± 0.03	0.53 ± 0.05	0.78 ± 0.02	0.77 ± 0.04	0.59 ± 0.05
TOHA (ours)	✓	**0.76 ± 0.04**	**0.60 ± 0.09**	0.89 ± 0.01	**0.96 ± 0.01**	0.66 ± 0.05
LLama-2-7B						
SelfCheckGPT [1]	✗	0.60 ± 0.04	**0.60 ± 0.03**	0.77 ± 0.02	0.78 ± 0.02	0.67 ± 0.04
Semantic entropy [2]	✗	0.56 ± 0.03	0.49 ± 0.03	0.79 ± 0.01	0.77 ± 0.02	0.63 ± 0.04
EigenScore [3]	✗	0.57 ± 0.04	0.52 ± 0.06	0.61 ± 0.03	0.78 ± 0.02	0.65 ± 0.03
TOHA (ours)	✓	**0.65 ± 0.02**	0.56 ± 0.02	**0.90 ± 0.01**	**0.87 ± 0.04**	**0.68 ± 0.05**
LLaMA-2-13B						
SelfCheckGPT [1]	✗	0.61 ± 0.05	**0.60 ± 0.06**	0.88 ± 0.02	0.75 ± 0.04	0.61 ± 0.04
Semantic entropy [2]	✗	0.60 ± 0.03	0.52 ± 0.03	0.77 ± 0.04	0.70 ± 0.02	0.62 ± 0.03
EigenScore [3]	✗	0.58 ± 0.04	0.48 ± 0.05	0.59 ± 0.03	0.60 ± 0.02	0.54 ± 0.05
TOHA (ours)	✓	**0.67 ± 0.04**	0.56 ± 0.05	**0.92 ± 0.02**	**0.88 ± 0.05**	**0.66 ± 0.03**

Table 6: ROC AUC (↑) of multiple generations-based methods (with 20 additional samples) and TOHA. The best results for each model are highlighted in **bold**, and the second best are underlined.

Method	Single gen.	CoQA	SQuAD	XSum
LLaMA-3.1-8B				
SelfCheckGPT [1]	✗	**0.95 ± 0.01**	0.78 ± 0.03	**0.75 ± 0.03**
Semantic entropy [2]	✗	0.82 ± 0.03	0.54 ± 0.06	0.46 ± 0.05
EigenScore [3]	✗	0.84 ± 0.01	0.56 ± 0.03	0.48 ± 0.03
TOHA (ours)	✓	0.84 ± 0.01	**0.87** ± 0.03	0.65 ± 0.05
Qwen2.5-7B				
SelfCheckGPT [1]	✗	0.75 ± 0.02	**0.77 ± 0.02**	**0.72 ± 0.03**
Semantic entropy [2]	✗	0.83 ± 0.05	0.58 ± 0.04	0.69 ± 0.05
EigenScore [3]	✗	0.83 ± 0.05	0.56 ± 0.03	0.48 ± 0.03
TOHA (ours)	✓	0.79 ± 0.05	**0.77 ± 0.02**	0.69 ± 0.03

Method	Llama-2-7B	Llama-2-13B
ReDEEP [8]	0.68 ± 0.02	0.77 ± 0.01
TOHA (ours)	**0.70 ± 0.04**	**0.80 ± 0.02**

Table 7: Performance comparison between ReDEEP and TOHA on the entire RAGTruth dataset.

Table 8: ROC-AUC values of supervised classifiers on top of various set of features. TOP-1 results are highlighted with **bold font**, while TOP-2 are underlined.

Features	MS MARCO	CoQA
Mistral-7B		
Standard topological	0.67	0.69
Sparsity ratio	0.66	0.7
Entropy	0.75	0.77
Wasserstein	0.77	0.73
Spectral norm	0.73	0.72
Attention to <s>	0.65	0.61
MTop-Div	**0.86**	**0.98**
LLaMA-2-7B		
Standard topological	0.69	0.7
Sparsity ratio	0.49	0.61
Entropy	0.38	0.68
Wasserstein	0.73	0.6
Spectral norm	0.49	0.64
Attention to <s>	0.62	0.64
MTop-Div	**0.75**	**0.96**

way to measure their similarity. Finally, we analyzed the average attention to the first token as we discovered that hallucination-aware heads often attend to in presence of hallucinations (see Section 4).

To identify the most informative features for hallucination detection, we trained $L1$-regularized supervised classifiers on concatenated features from all layers and heads and compared them to classifiers trained only on $\mathrm{MTop\text{-}Div}_G(R, P)$ values under the same conditions. The results are presented in Table 8.

While the classifier based on $\mathrm{MTop\text{-}Div}_G(R, P)$ values significantly outperforms alternative approaches, computing these values across all layers and attention heads is highly computationally expensive. To address this, we developed TOHA — a more efficient alternative that aggregates divergence values from only a subset of the "hallucination-aware" attention heads.

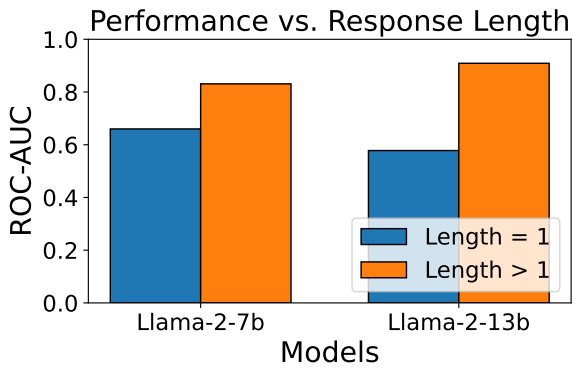


Figure 7: Comparison of ROC-AUC scores (↑) for single-word versus multi-word model responses. Dataset: SQuAD.

B.4 Other metrics for the head selection procedure

Additionally, we investigated alternative attention-map-based scores — including entropy, spectral norm, and the Wasserstein distance between the persistent diagrams of prompts and responses — for selecting specialized attention heads. Following the pipeline of the Algorithm 1, we computed the average distances between hallucinated and grounded samples using alternative scores. The results, presented in Figure 10, reveal that the heads that separate samples best for MS MARCO do not generalize to the CoQA dataset. This suggests that our proposed $\mathrm{MTop\text{-}Div}_G(R, P)$ metric is better suited for the task compared to existing solutions.

B.5 Performance analysis

Additionally, we analysed TOHA's performance for extremely short responses consisting of one word. The results are presented in Figure 7. While its performance drops — as expected when the response graph degenerates to a single vertex — it remains non-random. This indicates that TOHA retains some predictive power even for these challenging, out-of-design cases.

As for the TOHA's performance on longer responses, the results for the MS MARCO, CNN/DM and XSum datasets demonstrate that as the response length grows, the hallucination signals become less distinguishable for all the considered methods, as hallucinated tokens comprise a very small part of the response. However, TOHA remains superior to baselines, confirming that our topology-based approach is more effective than existing methods for complex long-form responses.

Probe set	CNN/DM	XSum
MS MARCO	0.272	1.0
CNN/DM	1.0	0.746
CoQA	0.499	0.134
SQuAD	0.373	0.346
XSum	0.515	1.0

Table 9: P-values of the T-test for the transferability experiment.

B.6 P-values for the transferability experiment

As we mentioned, for CNN/DM and Xsum dataset, the TOHA's performance in transfer settings fall within the standard deviation of the method. To confirm this, we calculated the p-vales of the corresponding T-test for means of two independent samples. The results, provided in Table 9, support our claim: all p-values are significantly greater than 0.05.

C Topological Data Analysis: Background

A simplicial complex S is a collection of simplices such that every face of a simplex $\sigma \in S$ is also in S. Simplices are the higher-dimensional generalizations of triangles; a 0-simplex is a vertex, a 1-simplex is an edge, a 2-simplex is a triangle, and so forth. Formally, given a finite set X, an n-simplex σ is an $(n+1)$ subset of X. Simplicial complexes are fundamental objects in algebraic and combinatorial topology, serving as a discrete analog to topological spaces.

The Vietoris-Rips complex $VR_\varepsilon(X)$ of a weighted graph $G = (V_G, E_G)$ with distance threshold $\varepsilon > 0$ is defined as follows:

$$\mathrm{VR}_\varepsilon(G) = \left\{ \sigma \subseteq V_G \;\middle|\; \forall\, v_i, v_j \in \sigma,\ w(e_{ij}) \leq \varepsilon \right\},$$

where w is the edge weight function associated with G.

Homology groups H_k are invariants used in algebraic topology to study the topological properties of a space. Let $C_k(S)$ denote vector space over $\mathbb{Z}/2\mathbb{Z}$, with the basis consisting of k-dimensional simplices of S. Elements of C_k are called chains. Formally, homology groups are derived from a chain complex $(C_\bullet, \partial_\bullet)$, which is a sequence of

C_k connected by boundary maps ∂_k:

$$C_\bullet : \cdots \to C_{k+1} \xrightarrow{\partial_{k+1}} C_k \xrightarrow{\partial_k} \cdots,$$
$$\partial_k \circ \partial_{k+1} = 0.$$

The k-th homology group H_k is defined as the quotient of the group of k-cycles (chains whose boundary is zero) by the group of k-boundaries (chains that are the boundary of a $(k+1)$-chain). Mathematically, this is expressed as:

$$H_k(S) = Z_k(S)/B_k(S),$$

where $Z_k = \ker \partial_k = \{c \in C_k \,|\, \partial_k(c) = 0\}$ and $B_k = \operatorname{im} \partial_{k+1} = \{\partial_{k+1}(c) \,|\, c \in C_{k+1}\}$ is the group of k-boundaries. The elements of $H_k(S)$ represent various k-dimensional topological features in S. Elements of a basis in $H_k(S)$ correspond to a set of basic topological features.

A filtration of simplicial complexes $\mathcal{F}$ is a family of nested simplicial complexes:

$$\mathcal{F} : \varnothing \subseteq S_1 \subseteq S_2 \subseteq \cdots \subseteq S_n = S,$$

where each S_k is a simplicial complex itself. In practice, the filtrations of simplicial complexes are usually obtained for sequences of increasing thresholds $0 < \varepsilon_1 < \cdots < \varepsilon_n$. For example, simplicial complexes $VR_{\varepsilon_i}(X)$ form a filtration

$$\mathcal{F}_{VR}(X) : \varnothing \subseteq VR_{\varepsilon_1}(X) \subseteq VR_{\varepsilon_2}(X) \subseteq \ldots$$
$$\subseteq VR_{\varepsilon_n}(X) = VR(X).$$

As the threshold ε increases, new topological features (e.g., connected components, holes) can appear and disappear. The persistent homology tool tracks the dynamics of these topological features. Formally, the k-th persistent homology of S is the pair of sets of vector spaces $\{H_k(S_i) \,|\, 0 \le i \le n\}$ and maps f_{ij}, where $f_{ij} : H_k(S_i) \to H_k(S_j)$ is a map induced by the embedding $S_i \subseteq S_j$. Each persistent homology class in this sequence is "born" at some S_i and "dies" at some S_j or never dies (Barannikov, 1994). This birth-death process of a basic set of independent topological features can be visualized as the set of intervals $[\varepsilon_{\text{birth}}, \varepsilon_{\text{death}}]$ called barcode (see Figure 8). The features with 0 lifespans are typically excluded. The horizontal axis is a sequence of thresholds ε, and each horizontal bar corresponds to a single feature. We begin with $|X| = m$ connected components (all of them are "born"), and as ε increases, their pairs are merged (each merge corresponds to a "death" of a feature). The $0-$th barcode construction procedure is equivalent to Kruskal's algorithm for minimum spanning tree (MST), the bars in the barcode correspond to the edges in the MST of X (Tulchinskii et al., 2023).

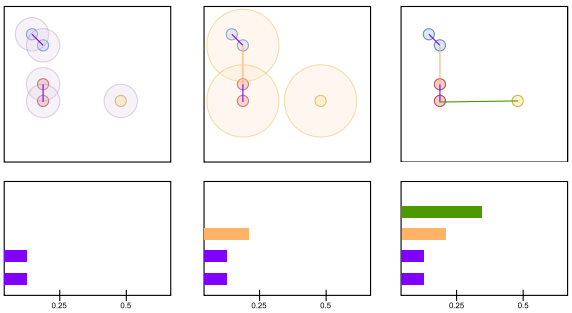


Figure 8: H_0 barcode construction. As the threshold increases, the separate connected components merge, resulting in the death of topological features. The horizontal axis is a sequence of thresholds ε, and each horizontal bar corresponds to a single feature.

D Properties of $\text{MTop-Div}_G(R, P)$

Proof of Proposition 3.1. The $0-$th Cross-Barcode coincides with the set of edges in the minimal spanning tree of the weighted graph G with all the weights within P vertex subset equal zero. Excluding the zero weight edges, this edge set coincides with the minimal spanning forest attaching the vertex set R to P vertices. □

Other properties of MTop-Div for attention graphs. Here we present some other properties of our proposed $\text{MTop-Div}_G(R, P)$.

Proposition D.1. *The following holds for any attention graph G with vertex set V_G and its complementary vertex subsets P, R, where $P \cup R = V_G$ and $P \cap R = \varnothing$.*

- *The divergence itself is bounded by*

$$0 \le \text{MTop-Div}_G(R, P) \le |R|. \quad (4)$$

- ***(Stability.)*** *If the weights of G change by no more than ε, then the corresponding $\text{MTop-Div}(R, P)$ changes by no more than $\delta = \varepsilon|R|$.*

- ***(Exact sequence.)*** *For any α, the following sequence of natural maps of homology groups is exact*

$$(\mathbb{Z}/2\mathbb{Z})^{|P|} \xrightarrow{r_2} H_0(VR_\alpha(G)) \xrightarrow{r_1}$$
$$\xrightarrow{r_1} H_0(VR_\alpha(G, w_{(R\cup P)/P})) \xrightarrow{r_0} 0.$$

- ***(Connection with hallucinations.)*** *The normalized divergence value* $\frac{1}{|R|}\,\text{MTop-Div}(R,P) = 0$ *iff the MSF attaches every response token to a prompt token by a subtree with attention weights* $= 1$.

Proof of Proposition D.1.
1. This property is immediately obtained from the properties of an attention map: all its weights lie between 0 and 1. □
2. Denote by $\text{MSF}(R,P)$ the minimum spanning forest attaching R to P. Note that we have properties D.1, so

$$\text{MTop-Div}(R,P) = \sum_{e \in \text{MSF}(R,P)} w(e). \quad (5)$$

Therefore, we have to show that the weight of $\text{MSF}(R,P)$ does not change significantly when all weights are changed by no more than ε.

There are two possibilities: 1) after a change, all MSF edges remain the same, or 2) some edges are replaced with other edges. In the first case, it is obvious that the total sum of edge weights changes by no more than

$$\delta = \varepsilon \cdot \#\text{edges}(\text{MSF}(R,P)) = \varepsilon \cdot |R|.$$

Consider the second case. Denote by MSF_{prev} the original MSF, by MSF_{new} — the MSF after the change; let w be the edge weight function before the change, $\hat{w}$ — after the change. The following inequalities hold:

$$\hat{w}(\text{MSF}_{\text{new}}) < \hat{w}(\text{MSF}_{\text{prev}}); \quad (6)$$

$$w(\text{MSF}_{\text{prev}}) - \delta \le \hat{w}(\text{MSF}_{\text{prev}}) \le \le w(\text{MSF}_{\text{prev}}) + \delta; \quad (7)$$

$$w(\text{MSF}_{\text{new}}) - \delta \le \hat{w}(\text{MSF}_{\text{new}}) \le \le w(\text{MSF}_{\text{new}}) + \delta; \quad (8)$$

$$w(\text{MSF}_{\text{new}}) \ge w(\text{MSF}_{\text{prev}}). \quad (9)$$

From (6)-(7) follows that $\hat{w}(\text{MSF}_{\text{new}}) < w(\text{MSF}_{\text{prev}}) + \delta$; from (8)-(9) follows that $\hat{w}(\text{MSF}_{\text{new}}) \ge w(\text{MSF}_{\text{prev}}) - \delta$. □
3. We have to check the definition of the exact sequence: $\text{Ker}(r_i) = \text{Im}(r_{i+1})$. For a pair r_0, r_1, it is equivalent to the surjectvity of r_1. The H_0 homology group of a graph corresponds to the connected components of the graph. The set of edges $E^{\le\alpha}_{(G,w)} = \{e \in E_G | w_e \le \alpha\}$ is always a subset in the analogous set of the weighted graph $(G, w_{(R\cup P)/P})$ with all weight edges between P vertices set to zero. Therefore, the map r_1 between their connected components is surjective. Similarly, the kernel of the map r_1 is spanned by the differences of two connected components, which are merged after adding some of the edges between P vertices, and any such difference lies in the image of the map r_2. Also, any two vertices from P belong to the same connected component in the graph $(G, w_{(R\cup P)/P} \le \alpha)$, hence the image of r_2 is in the kernel of r_1. Therefore, the considered sequence is exact indeed. □4. Follows obviously from the MSF formula for $\text{MTop-Div}(R,P)$ and attention map properties. □

Intuition behind MTop-Div properties and hallucination detection. The stability property guarantees that similar attention patterns yield similar hallucination scores, making the metric's behavior consistent and predictable. The exact sequence property formalizes geometric intuition behind our metric — it measures the strength of the response's connection to the prompt through multiscale topological features of the attention graph. In the last property, we present an "ideal" case: if a model "knows what to look at" — each token in the response attends to some token in the prompt with an attention weight equal to 1 — MTop-Div would be equal to 0, indicating zero uncertainty.

E Datasets

SQuAD (Rajpurkar et al., 2016) and CoQA (Reddy et al., 2019) are widely used English question-answering benchmarks that have facilitated the development of hallucination detection datasets (Kuhn et al., 2023; Manakul et al., 2024). Similarly, XSum (Narayan et al., 2018), a dataset of news articles with one-sentence summaries, is commonly employed in hallucination detection research for abstractive summarization (Shi et al., 2024; Cao et al., 2022). To assess LLM performance, we used GPT-4o to annotate responses to questions sourced from SQuAD, CoQA, and summarization tasks from XSum.

E.1 Data Generation & Annotation

Generation. We generate responses from a language model (LLM) for the considered datasets, employing different prompting strategies for each dataset while keeping these strategies consistent across models (see prompt examples in Table 10). For SQuAD and XSum, responses are generated using a zero-shot approach. In contrast, for CoQA,

SQuAD	CoQA
Given the context, answer the question in a brief but complete sentence. Note that your answer should be strictly based on the given context. In case the context does not contain the necessary information to answer the question, please reply with "Unable to answer based on given context". *Context:* Once upon a time, in a quiet village, there lived a kind old baker named Henry. He was known for his delicious bread and warm smile. One day, a traveler arrived, tired and hungry, and Henry welcomed him with a fresh loaf. *Question:* Who was known for baking delicious bread? *Answer:*	Once upon a time, in a quiet village, there lived a kind old baker named Henry. He was known for his delicious bread and warm smile. One day, a traveler arrived, tired and hungry, Henry welcomed him with a fresh loaf. *Q:* What was Henry known for? *A:* Baking delicious bread. *Q:* What else? *A:* Warm smile. *Q:* How did the traveler feel when he arrived? *A:* Tired and hungry. *Q:* What did Henry give the traveler?

Table 10: Examples of prompts used during generation for CoQA and SQuAD (we add additional delimiter spaces and formatting not present in actual prompts for better readability). SQuAD contains instructions followed by context and questions. In CoQA, the prompt has only a contextual passage followed by a question-and-answer series, with the last question being the actual one.

XSum
Please annotate potentially hallucinated model-generated summaries in the following settings. I will provide a reference text and a model-generated summary of this text. You will judge whether the given model-generated summary contains hallucinations. Answer "Yes" if the summary contains hallucinations, "No" if it does not, and "N/A" if you cannot decide. Do NOT give any extra explanations.

Table 11: The prompt used during generation for the XSum dataset (we add additional delimiter spaces and formatting not present in actual prompts for better readability).

You are an AI assistant specialized in detecting hallucinations in question-answering tasks.
Your job is to analyze the given context, question, and generated answer to identify whether the answer contains any hallucinations. Examples:

Example 1.
Context:
The city of Paris is the capital of France. It is known for its iconic landmarks like the Eiffel Tower and Notre Dame Cathedral.
The city is situated in the northern part of the country, near the Seine River.
Question: Is Paris the capital of Germany?
Generated answer: Yes, Paris is the capital of Germany.
Hallucination: Yes.

Example 2.
Context:
The city of Paris is the capital of France.
It is known for its iconic landmarks like the Eiffel Tower and Notre Dame Cathedral.
The city is situated in the northern part of the country, near the Seine River.
Question: Is Paris the capital of Germany?
Generated answer: No, Paris is not the capital of Germany. According to the context, Paris is the capital of France.
Hallucination: No.

You should determine if the answer contains hallucinations according to the hallucination types above.
If you cannot decide if the generated answer is a hallucination, write "N/A." as the answer.
The answer you give MUST be ONLY "Yes.", "No." or "N/A."; do NOT give ANY explanation.

Table 12: Example of annotation prompt passed to GPT-4o (we add additional delimiter spaces and formatting not present in actual prompts for better readability).

Prompt number		1	2	3	4	5	Average
CoQA	Accuracy (↑)	0.809 ± 0.017	0.861 ± 0.015	0.742 ± 0.003	0.795 ± 0.009	0.831 ± 0.025	0.808
	Precision (↑)	0.849 ± 0.021	0.911 ± 0.007	0.771 ± 0.003	0.828 ± 0.011	0.860 ± 0.012	0.844
	Recall (↑)	0.871 ± 0.004	0.877 ± 0.019	0.877 ± 0.013	0.877 ± 0.005	0.893 ± 0.027	0.879
SQuAD	Accuracy (↑)	0.831 ± 0.003	0.857 ± 0.018	0.857 ± 0.008	0.872 ± 0.003	0.854 ± 0.007	0.854
	Precision (↑)	0.813 ± 0.002	0.831 ± 0.028	0.845 ± 0.021	0.850 ± 0.011	0.847 ± 0.007	0.837
	Recall (↑)	0.796 ± 0.008	0.839 ± 0.010	0.823 ± 0.023	0.858 ± 0.018	0.813 ± 0.017	0.826

Table 13: Classification metrics of GPT-4o annotation for CoQA and SQuAD with human labels considered as true. The table shows metric scores for different variants of prompts used and the score value averaged across all prompt variants.

Pair	Accuracy	Pearson-r
1 vs 2	0.783	0.6
1 vs 3	0.812	0.631
2 vs 3	0.826	0.653
Mean	0.807	0.628
Std	0.022	0.027

Table 14: Cross-annotator consistency metrics.

we create queries in a few-shot manner without providing specific instructions, following (Lin et al., 2024): each sample consists of a passage and a series of question-answer pairs, concluding with a final question that the model is expected to answer.

Annotation: automated vs human. We treat hallucination detection as a binary classification problem; our target indicates whether a hallucination is present anywhere in the model's response. Two approaches to annotating model generations were considered: 1) automated annotation using an LLM (in our case, GPT-4o), and 2) manual annotation by human experts.

During the automated annotation process, we provide an LLM's output preceded by an instruction (prompt) to GPT-4o. In this prompt, GPT-4o is asked to determine whether the output contains hallucinations, and we expect a single-word response of either "Yes" or "No." An example of such an instruction for the question answering task is shown in Table 12.

For human annotation, we asked three team members with at least upper-intermediate English proficiency to independently annotate approximately 100 samples from each dataset. The human annotation consistency metrics are provided in Table 14. We selected samples where all annotators reached a consensus and considered these annotations the ground truth hallucination labels.

To further evaluate GPT-4o, we conducted automatic annotation using several variations of prompts, each reformulating the task for GPT-4o, including zero-shot and few-shot versions. We then compared these annotations to the actual hallucination labels. The results, presented in Table 13, demonstrate a consistent alignment between GPT-4o's annotations and those made by humans, regardless of the specific prompt. This consistency confirms the robustness of our approach to the exact form of instruction.

Based on these findings, we prefer automated annotation as a cost-effective and efficient alternative to human experts.

Annotation: general pipeline. CoQA and SQuAD contain questions paired with ground-truth answers. To minimize false positives in labeling, we employed a two-step verification process:

1. Rouge-L scoring: we computed Rouge-L scores (using the `evaluate` library, v0.4.6) between the model's response and the ground-truth answers.

2. Substring matching: we checked whether any ground-truth answer was a substring of the response.

Responses with a Rouge-L score of 1 (exact match) were labeled as grounded. Those meeting both of the following criteria were flagged as potential hallucinations:

- Rouge-L score ≤ 0.3 (following (Kuhn et al., 2023));
- no ground-truth answer appears as a substring.

These candidate hallucinations were then reviewed by GPT-4o, and only confirmed cases were finally labeled as hallucinations.

For XSum, where reference summaries are more complex than the ground truth answers in SQuAD/CoQA, we bypassed Rouge-L filtering and relied solely on GPT-4o for annotation.

Detailed statistics for each dataset can be seen in Table 15. The number of samples in the datasets

Model	CoQA		SQuAD		XSum	
	Hal.	Grounded	Hal.	Grounded	Hal.	Grounded
Mistral-7B	776	776	311	389	301	448
LLaMA-2-7B	375	375	440	258	239	507
LLaMA-2-13B	279	384	314	436	208	522
LLaMA-3.1-8B	356	350	350	400	243	407
Qwen2.5-7B	171	218	423	423	194	556

Table 15: Datasets statistics. Number of hallucinated and grounded samples of each model.

varies across models, as we tried to maintain a balance of hallucinated and grounded responses, ensure sample cleanness, and minimize mislabeling. The procedure outlined above selects a different number of objects in a sample depending on the quality of the model's responses. Additionally, we provide the statistics regarding the response length distribution for each of the datasets we gathered (Figure 9).

F Implementation details

In this section, we describe the key implementation choices.

- For EigenScore, we used the last token representation to embed sentences, as suggested in (Chen et al., 2024). We took outputs from the 16th layer, since middle layers were shown to contain the most factual information (Sky et al., 2024; Azaria and Mitchell, 2023).
- For SelfCheckGPT, we used its NLI-based version.
- For LLM-Check, we considered its white-box attention score modification, as it works in a setting similar to ours.
- The topological divergences were calculated using `ripser` library (Tralie et al., 2018), MIT license.
- For the RAGTruth dataset, the model settings were aligned with the original paper.

All experiments were carried out using NVidia L40.

G Use of scientific artifacts & AI assistants

CoQA contains passages from seven domains under the following licenses: Literature and Wikipedia passages are shared under CC BY-SA 4.0 license; Children's stories are collected from MCTest which comes with MSR-LA license; Middle/High school exam passages are collected from RACE which comes with its own license; News passages are collected from the DeepMind CNN dataset which comes with Apache license. SQuAD dataset comes under CC BY-SA 4.0 license. RAGTruth dataset comes under MIT license. XSum dataset comes under MIT license.

We used all the artifacts as it was intended by the corresponding licenses. No personal information or offensive content is contained in the considered datasets.

The original text of this paper was spell- and grammar-checked and slightly smoothed using Grammarly.

H Potential risks

1. Ethical risks from deployment: overconfidence in TOHA's scores could lead to unchecked LLM outputs in high-stakes scenarios (e.g., healthcare). TOHA should be frame as a "warning system" rather than a definitive filter, and advocate for human review.
2. Attention manipulation attacks: adversarial prompts could artificially alter attention patterns, evading detection.

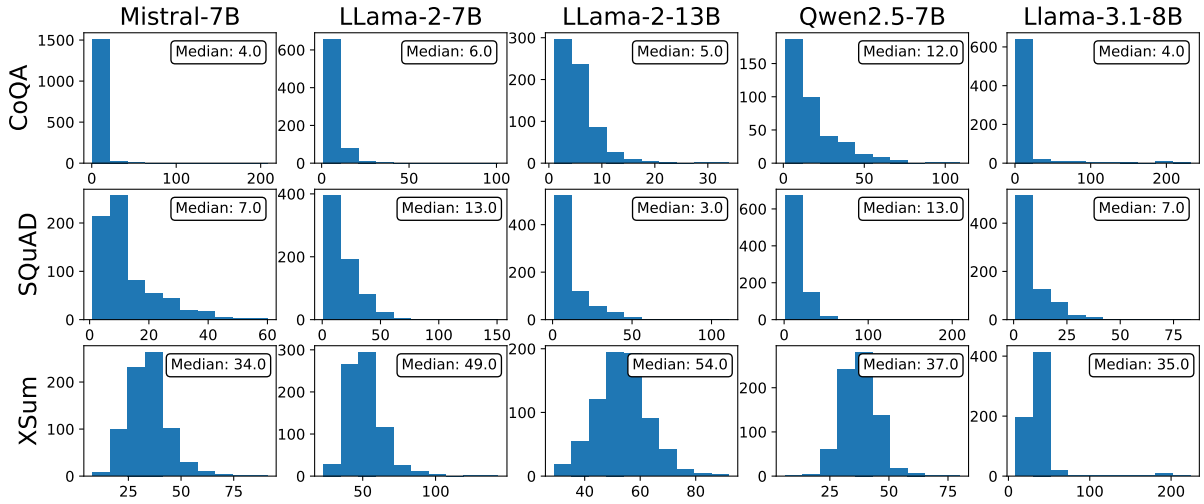


Figure 9: The distributions of response lengths (in words).

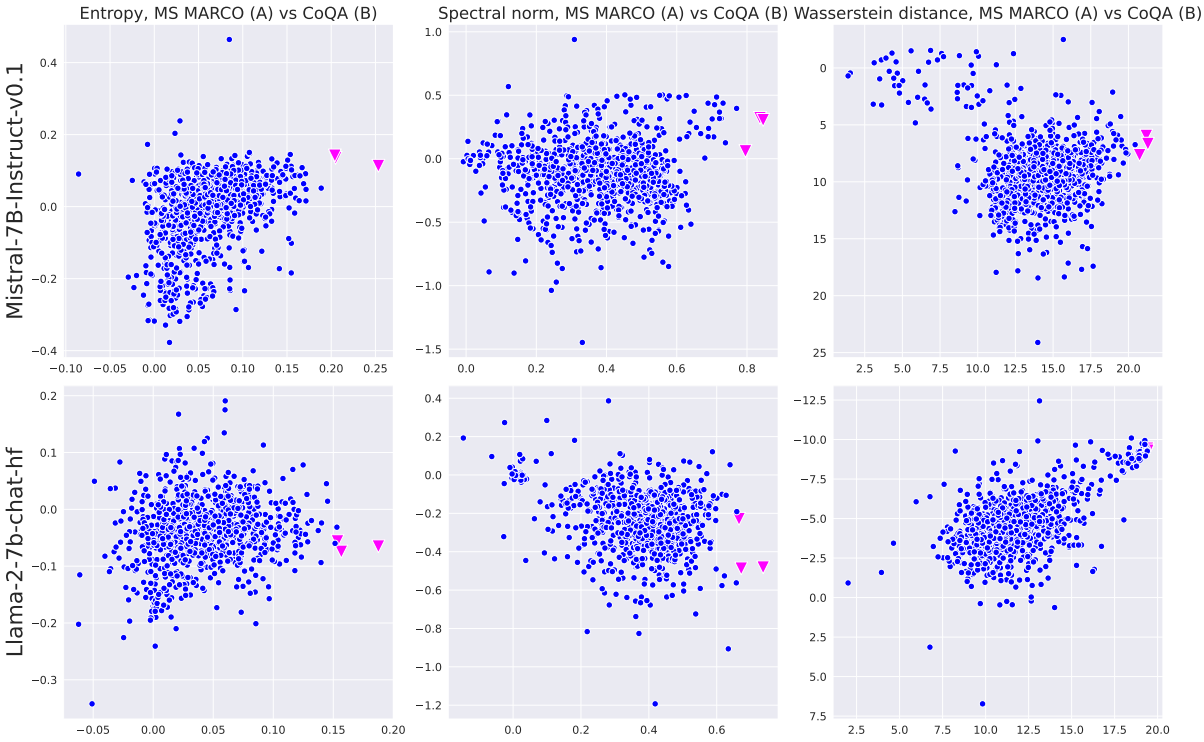


Figure 10: Δ_{ij} values for ij heads, MS MARCO and CoQA. Vertical axis corresponds to the difference on the dataset (B), horizontal to the difference on the dataset (A). The heads that separate samples best are highlighted in pink.